

# Sameer Verma

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## EDUCATION

### University of Cincinnati

Cincinnati, OH

BS in Mechanical Engineering, Minor in Mathematics GPA: 3.36/4.0

Expected Graduation: May 2029

- Relevant Courses: Thermodynamics, Solid Mechanics, Dynamics for Engineering, Circuits & Sensing, Material Science, Differential Equations, Physics 1&2, Design Graphics

## WORK EXPERIENCE

### Gas Dynamics and Propulsion Laboratory

Cincinnati, OH

University of Cincinnati

May 2026 – August 2026 | Full-time | 40 hrs/week

September 2026 – Present | Part-time | 10 hrs/week

- Engineered a low-cost (~\$140) custom strain measurement system for GE Aerospace-funded rotating detonation engine research, reducing instrumentation cost by ~98% versus a ~\$10,000 conventional system.
- Performed modal FEA in SolidWorks to predict combustor natural frequencies and identify expected structural mode locations in the frequency domain.
- Designed, assembled, and soldered a thermocouple signal-conditioning circuit using an AD8421 instrumentation amplifier, providing ~166x gain (41  $\mu\text{V}/^\circ\text{C}$  to 6.8  $\text{mV}/^\circ\text{C}$ ) for combustion diagnostics.
- Built a Python pipeline to convert 1 MHz strain-gauge data to microstrain, apply a Chebyshev Type II low-pass filter, and perform time-frequency spectrogram analysis to interpret combustor structural frequencies and correlate strain response with chamber pressure during post-test review of RDE firing data.
- Diagnosed and mitigated signal-to-noise and EMI issues in the strain measurement chain through iterative improvements to shielding, grounding, amplifier placement, wiring, and high-temperature gauge bonding.

### Guerbet

Cincinnati, OH

Manufacturing & Quality Intern

August 2025 – December 2025 | Full-time | 40 hrs/week

- Tested and validated OptiStar injector assemblies, troubleshooting zeroing calibration and volume discrepancy issues to ensure proper operation.
- Identified and investigated a software-driven volume discrepancy issue using LabVIEW, helping engineers analyze root causes faster by 50% and confirm correct mechanical–software interaction during validation.

## LEADERSHIP EXPERIENCE & PROJECTS

### UC Rocketry Club

Cincinnati, OH

Student Engineer

January 2026 – Present

- Designed a custom battery holder for the rocket's avionics bay using Siemens NX, producing CAD models and manufacturing drawings for flight-hardware integration.
- Used OpenRocket to analyze flight performance, trajectory, and stability throughout design iterations.
- Participated in manufacturing, assembly, testing, and launch preparation of a two-stage rocket, supporting launch operations and evaluation of the completed vehicle.
- Collaborated with a multidisciplinary team of ~10–15 engineers on design reviews, manufacturing, and testing.

### NASA L'Space NPWEE (Team 18)

Tempe, AZ (Remote)

Mechanical Design Lead

May 2026 – August 2026

- Co-authored a NASA proposal investigating plume-surface interactions (PSI) between single-, dual-, and quad-nozzle lunar lander engine configurations using BP-1 regolith simulant, as part of NASA's L'Space Proposal Writing and Evaluation Experience.
- Led SolidWorks design of supersonic cold-flow converging-diverging nozzles and swappable acrylic mounting plates for each nozzle configuration, plus a 3D-printed PETG regolith tray, for an in-house-fabricated compressed-air plume-impingement test campaign.
- Contributed to a 12-month NASA Phase B–F technology development plan and \$6,700 project budget, culminating in a quad chart and New Technology Report (NTR) submission.

### Autonomous Multipurpose UAV Drone

Cincinnati, OH

Personal Project

April 2026 – Present

- Lead mechanical architecture and structural design of a modular UAV airframe in Fusion 360, currently redesigning the fuselage around a removable, serviceable avionics bay.
- Drive rapid design iteration through PETG prototyping, applying design-for-additive-manufacturing principles and heat-set threaded inserts to create durable, serviceable mechanical joints.
- Own mechanical packaging and integration of propulsion, batteries, electronics, and onboard computing, balancing structural integrity, component retention, cooling, accessibility, and center of gravity.

## SKILLS

**Technical:** Siemens NX, SolidWorks, MATLAB, Python, Java, LabVIEW, VBA, Microsoft Office, Fusion 360, Soldering, Circuit Prototyping, 3D Printing / Additive Manufacturing